

AI Full Stack Developer Assignment

PDF Knowledge Base AI Chatbot (RAG System)

Objective

Build an AI-powered Knowledge Base Chatbot where an administrator can upload PDF documents that become the AI's knowledge base. Users can ask questions through a public chatbot interface, and the AI should answer questions using the uploaded documents while suggesting relevant follow-up questions.

The project must use a microservice architecture with a separate Node.js backend and Python AI service communicating through Redis Pub/Sub.

Mandatory Tech Stack

Frontend

- Next.js (App Router)
- TypeScript
- Tailwind CSS
- SHADCN UI

Backend

- Node.js
- NestJS or Express
- TypeScript

AI Service

- Python
- FastAPI
- LangChain

- LangGraph

Database

- PostgreSQL or MongoDB

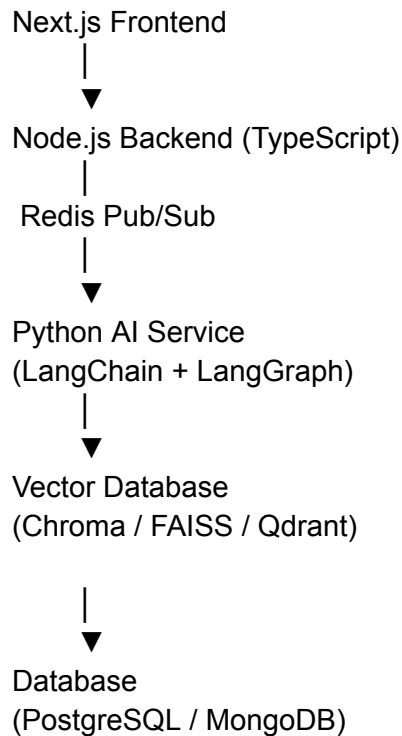
Vector Database (Free)

- ChromaDB (Preferred)
- OR FAISS
- OR Qdrant

Communication

- Redis Pub/Sub (Mandatory)
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System Architecture



Module 1 — Admin Panel

Authentication

- Admin Login
 - Secure Authentication
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Dashboard

Display:

- Total Uploaded PDFs
 - Total Chat Sessions
 - Total Questions Asked
 - Recent Uploaded Documents
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Knowledge Base Management

Admin should be able to:

- Upload PDF files
- View uploaded PDFs
- Search uploaded PDFs
- Delete PDFs
- Reprocess PDFs

After uploading a PDF, the system should automatically:

- Extract text
 - Split into chunks
 - Generate embeddings
 - Store vectors in the Vector Database
 - Save metadata in the Database
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Module 2 — Public AI Chat

No authentication required.

Create a ChatGPT-style chatbot interface.

Features:

- Ask questions
 - Streaming responses
 - Markdown rendering
 - Typing indicator
 - Responsive UI
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AI Responses

Each response should display:

- AI Answer
 - Source Document Name
 - Page Number (if available)
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Suggested Follow-up Questions (Mandatory)

After every response, automatically generate **3–5 relevant follow-up questions** based on the conversation and retrieved knowledge.

Example:

Suggested Questions

- What happens if I forget my password?
 - How can I update my email?
 - Where can I change my profile settings?
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Conversation Memory

The chatbot should remember the current conversation so follow-up questions work naturally.

Python AI Service

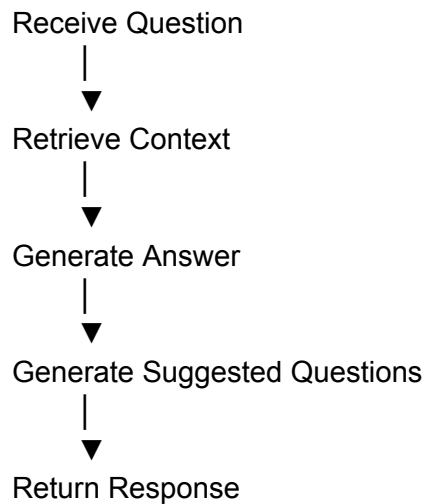
The AI service must be responsible for:

- PDF Processing
- Document Chunking
- Embedding Generation
- Vector Search
- RAG Pipeline
- Answer Generation
- Suggested Question Generation

Must use:

- LangChain
 - LangGraph
 - FastAPI
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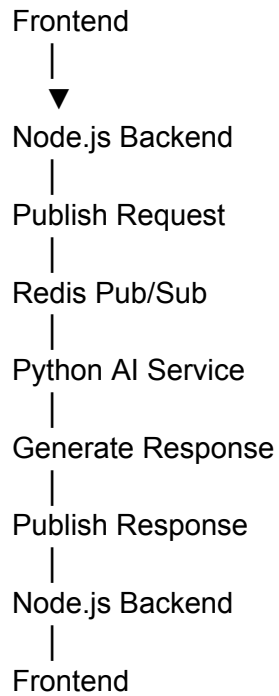
LangGraph Workflow (Mandatory)



Redis Pub/Sub Communication (Mandatory)

The Node.js backend and Python AI service **must communicate using Redis Pub/Sub**.

Expected flow:



Direct communication between the Node.js backend and Python AI service for AI processing is **not allowed**.

Database Design

Minimum collections/tables:

Users

- id
- email
- password

Documents

- id
- file name

- upload date
- processing status

Chats

- id
 - session id
 - question
 - answer
 - timestamp
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Required APIs

Admin APIs

- Login
- Upload PDF
- List PDFs
- Delete PDF
- Reprocess PDF

Chat APIs

- Ask Question
 - Chat History
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Project Structure

frontend/
backend/
python-ai/
shared/
README.md
docker-compose.yml (Optional)

Evaluation Criteria (100 Marks)

Criteria	Marks
Next.js Frontend	15
Node.js Backend	15
Python AI Service	20
LangChain Implementation	15
LangGraph Implementation	15
Redis Pub/Sub Integration	10
Database & Vector DB Design	5
Code Quality & Folder Structure	5

Submission Requirements

Candidates must submit the following:

- Public GitHub Repository
 - Complete source code
 - README with setup instructions
 - `.env.example`
 - Database schema
 - Architecture diagram
 - API documentation
 - **A public Loom or YouTube video (5–10 minutes) demonstrating:**
 - Project overview
 - Folder structure
 - Codebase walkthrough
 - System architecture
 - Redis communication
 - LangGraph workflow
 - PDF upload process
 - **AI chatbot working with follow up questions showing pdfs uploaded**
 - Suggested questions feature
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Important Notes

1. The project must be developed using **TypeScript** for the frontend and Node.js backend.
 2. The AI service **must be developed in Python** using **LangChain** and **LangGraph**.
 3. **Redis Pub/Sub is mandatory** for communication between the backend and AI service.
 4. A **free vector database** must be used.
 5. The chatbot must answer questions using the uploaded PDF knowledge base.
 6. Suggested follow-up questions are mandatory.
 7. The project should follow clean architecture and best coding practices.
 8. Proper error handling and validation should be implemented.
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Submission Deadline

Duration: 4 Days (TILL 6 August 12pm) from the date of assignment.

Failure to submit all required deliverables, including the public GitHub repository and demonstration video, may result in disqualification from the evaluation process.